Financial Intelligence & Advanced Analytics Platform

Client: Busy Insurance and Investment
Role Perspective: Principal AI Architect (AI Platforms & Distributed Systems)
Industry: BFSI / Insurance Operations
Duration: November 2025 - Present
Program Team Size: 19 cross-functional members

I designed and governed a finance-grade analytics and reporting platform that turned fragmented insurance operational data into a trusted decision layer for statutory reporting, actuarial analysis, management reporting, and CFO-level financial intelligence consumption.

Problem Context

The business was dealing with a familiar but painful problem: important data lived in too many places, and none of those places agreed cleanly with one another. Policy, claims, reinsurance, finance operations, and actuarial systems each carried their own definitions, their own timing, and sometimes their own version of the truth, which made reconciliation slow and reporting fragile.

The real challenge was not just moving data around. It was building a finance-traceable analytical foundation where SSIS, SQL Server, SSAS, and Power BI could work together under deterministic reconciliation rules, controlled lineage, and audit-ready governance. In a regulated environment, that matters more than flashy tooling ever will.

The platform also had to deal with inconsistent schemas, delayed extracts, messy source keys, and competing definitions for premium, loss, reserve, commission, earned exposure, and product hierarchy. None of that could be allowed to derail period close or erode trust in the outputs.

What Was Built

This platform established a 360-degree financial analytics and reporting platform for insurance operations, with architecture ownership spanning enterprise data model standardization, SSIS-driven ingestion control, SQL Server and SSAS analytical serving, Power BI reporting, production governance, and release readiness for regulated workloads.

I treated it as a controlled financial data platform, not a conventional BI build. That meant clear ownership boundaries for source stewardship, reconciliation governance, quality escalation, and reporting sign-off, so the platform could stand up under CFO, actuarial, and audit scrutiny without improvisation.

From an architectural point of view, the platform was designed for deterministic finance reporting rather than exploratory data-lake behavior. The focus was on governed dimensional conformance, auditable transformation layers, semantic consistency, traceable refresh orchestration, and role-based access control because those are the things that actually hold a BFSI reporting program together.

Business Goals

The goal was to create one governed financial intelligence platform that could support executive management reporting, actuarial analysis, statutory and audit-aligned reporting, and day-to-day finance transparency without forcing teams back into spreadsheets every time something needed to be reconciled.

There was also a quieter but equally important goal: reduce dependency on tribal knowledge. Metric definitions, lineage, and sign-off authority needed to live inside the platform itself instead of being scattered across people, email threads, and ad hoc reporting habits.

At the program level, the investment case was straightforward. Shorten reporting cycles, reduce reconciliation exceptions, standardize KPI interpretation across finance and actuarial teams, and tighten production discipline for anything that touched regulated reporting paths.

Platform Scope

The platform serves insurance finance, controllership, actuarial, statutory reporting, and audit stakeholders through a unified analytical backbone fed from policy, claims, commission, ledger, and related operational sources.

Core workloads include:

- Source-to-canonical ETL using SSIS with dependency-aware orchestration and restartability.
- Finance-grade storage and reconciliation layers in Microsoft SQL Server for conformed analytical consumption.
- SSAS Tabular semantic modeling for governed measures, time intelligence, and reusable financial dimensions.
- Power BI delivery for executive, operational finance, and actuarial analytics under governed row-level and report-level access controls.
- Production readiness, defect triage, and release sign-off workflows for controlled reporting deployments.

Architecture Choice

Azure was the right target for this program because the stack was already deeply aligned with Microsoft technologies: SSIS, SQL Server, SSAS Tabular, and Power BI were the operational core. Moving that workload to AWS or GCP would have introduced unnecessary translation work without changing the real center of gravity.

Azure also fits the control expectations of BFSI much better when paired with Entra ID, Key Vault, Private Link, Defender for Cloud, Monitor, Log Analytics, backup policies, and region-paired recovery. For a program shaped by CFO, actuarial, and audit stakeholders, the question is not “which cloud has the most features,” but “which one supports auditability, operational stability, and low-risk modernization best.” Here, Azure clearly did.

System Design

The architecture separates ingestion, curation, semantic serving, reporting, and governance so that a failure in one layer does not automatically collapse the entire reporting chain. That separation also makes ownership much clearer across source stewards, ETL teams, platform engineering, semantic modelers, BI developers, QA, and compliance reviewers.

A warehouse and semantic layer were chosen instead of direct source-to-dashboard federation. Finance-grade reporting needs repeatability, reconciliation checkpoints, and measure governance, and direct federation would have created more ambiguity, more duplicated logic, and weaker audit traceability.

At the lower level, the platform was broken into operationally isolated pieces: ingestion jobs, validation routines, conformance steps, dimensional publication, tabular processing, BI refresh orchestration, and monitoring. That structure reduced the blast radius of failures and made controlled restart possible at the package, table, model, or report-refresh boundary.

Security & Compliance Implementations

The security model was built around least privilege, private network reachability, auditable elevation, encryption in transit and at rest, and clear separation of duties between engineering, QA, business validation, and production release authority.

Key controls included:

- Entra ID-backed SSO with MFA, conditional access, and group-based entitlement mapping for Power BI, SQL, and admin access.

- Private connectivity for Azure SQL Managed Instance, Key Vault, storage endpoints, and integration runtimes through VNet isolation and Private Link.

- Transparent Data Encryption for SQL workloads, TLS for service-to-service communication, and secret rotation through Key Vault-backed references.

- Separation of duties across ETL development, semantic modeling, QA certification, and production deployment authority.

- Row-level security and dataset scoping in Power BI for entity, region, or function-specific access segmentation.

- Immutable audit logging for ETL runs, model refreshes, report publication, admin actions, and sign-off events.

- Data retention and evidence preservation aligned to statutory audit windows and internal governance requirements.

- Release gating that blocks uncertified datasets or reports from entering regulated consumption paths.

Cost Optimization Techniques

Cost control in this program is primarily a function of compute scheduling, storage tiering, semantic model sizing discipline, incremental processing, and reduction of avoidable reprocessing rather than GPU or token economics. Because the workload is batch-heavy and reporting-centric, the most meaningful optimizations come from architectural discipline in refresh boundaries, partition design, and operational automation.

Key cost techniques included:

- Using incremental SSIS loads and watermark-driven extraction to avoid full reloads for stable historical periods.
- Partitioning large fact tables and SSAS tabular models by fiscal period so that cube processing touches only changed partitions.
- Archiving immutable landing files and aged staging datasets into lower-cost object storage instead of preserving all historical operational data in hot SQL tiers.
- Right-sizing Azure SQL Managed Instance and analytical serving tiers based on close-cycle peaks versus business-as-usual usage, with non-production environments scheduled down outside working windows where feasible.
- Reusing governed semantic measures in SSAS and Power BI datasets to reduce duplicated computation and report-specific model sprawl.
- Enforcing report rationalization so unused or redundant dashboards do not continue driving unnecessary dataset refresh, support overhead, and semantic duplication.
- Automating failure detection and restart from checkpointed stages to minimize manual recovery effort and lost engineering time during ETL disruptions.

Core Tech Stack and Tools

Source Integration

- SSIS for controlled batch ingestion, source-specific extraction, transformation routines, file ingestion, and restartable ETL execution.
- Azure Data Factory with SSIS Integration Runtime for orchestration, scheduling, promotion, parameterization, and hybrid connectivity.
- SFTP connectors, flat-file ingestion utilities, and database-native connectors for policy, claims, commission, reinsurance, and ledger feeds.
- SQL Server Agent or ADF trigger chaining for package sequencing and close-window control.

Data And Storage

- Azure SQL Managed Instance for staging, conformance, reconciled warehouse serving, control tables, exception stores, and historical snapshots.
- SQL Server stored procedures, partitioning, indexed views where appropriate, and controlled T-SQL transformation logic.
- Azure Blob Storage for landing-zone archival, replayable retention, audit evidence preservation, and cold-path cost optimization.
- Backup vaults, geo-backup, long-term retention, and failover groups for resilience.

Semantic And Analytical Layer

- SSAS Tabular or Azure Analysis Services for enterprise semantic modeling, KPI reuse, period-aware partitions, and controlled publication of business semantics.
- Tabular Editor for model governance and deployment discipline.
- DAX Studio and SQL traces for performance tuning and bottleneck analysis.

BI Consumption

- Power BI Service for governed dataset publication, app workspaces, executive reporting, scheduled refresh, deployment pipelines, and stakeholder distribution.
- Power BI Desktop for authoring, model validation, and performance testing.
- Power BI Gateway where hybrid connectivity is still required.

- Power BI Premium or Fabric capacity only when scale justifies it.

DevOps And Release Control

- Azure DevOps Repos and Pipelines for ETL packaging, SQL release scripts, semantic model promotion, Power BI artifact lifecycle control, and gated approvals.

- Git-based branching with environment-specific parameterization.

- Controlled release checklists, regression packs, and sign-off workflows aligned to finance and audit boundaries.

Monitoring And Support

- Azure Monitor and Log Analytics for telemetry, diagnostics, failure alerts, refresh monitoring, and central evidence capture.

- SQL monitoring baselines for CPU, IO, tempdb pressure, blocking, execution hotspots, and ETL window behavior.

- Power BI refresh history, gateway telemetry, and dataset diagnostics.

- ServiceNow or equivalent ITSM tooling for incidents, defects, approvals, and support audit trails.

Identity And Compliance

- Microsoft Entra ID for SSO, group-based authorization, and environment-bound admin access.

- Azure Key Vault for connection strings, credentials, and secrets.

- Privileged Identity Management for just-in-time elevation.

- Defender for Cloud, SQL auditing, and immutable audit retention.

- Purview or similar lineage tooling if enterprise metadata governance is included in the broader trajectory.

Azure Resources and Hardware Used

Component	Suggested Azure Resource	Baseline Sizing

ETL Orchestration	Azure Data Factory with SSIS Integration Runtime	4-8 vCores equivalent for steady-state ETL, burstable during close windows.
Landing Archive	Azure Blob Storage	2-5 TB initial hot/cool mix depending on extract retention and replay policy.
Staging Database	Azure SQL Managed Instance	General Purpose, 8-16 vCores, 1-2 TB storage for standardized staging and operational control tables.

Warehouse Database	Azure SQL Managed Instance	Business Critical or General Purpose high-memory profile, 16-32 vCores, 2-4 TB storage depending on period history and partitioning strategy.
Semantic Layer	Azure Analysis Services / SSAS on Azure VM	S2-S4 equivalent or VM-backed tabular deployment with 64-128 GB RAM for finance models with period partitions.
BI Consumption	Power BI Premium / Fabric capacity where justified	Sized to refresh frequency, dataset memory footprint, and executive concurrency.
Monitoring	Log Analytics Workspace	90-365 day retention depending on audit and operational evidence requirements.

Security	Key Vault, Defender for Cloud, Private Endpoints	Standard enterprise deployment across all data services.
DR	Geo-backup / failover group / secondary region replicas	RPO/RTO selected against reporting criticality and close-cycle tolerance.

Impactful Outcomes Achieved

Metric	Baseline	Achieved
Reporting subject areas standardized	0 governed cross-domain model	6 major financial subject areas unified
Critical source systems onboarded	2 partially reconciled feeds	7 controlled source domains
Monthly reporting cycle preparation effort	~8-10 business days with heavy manual reconciliation	Reduced to ~4-5 business days for governed data preparation

Reconciliation exceptions requiring manual intervention	100% manually collated exception analysis	55-65% auto-classified through validation and exception tables
Executive dashboard refresh latency after warehouse publish	4-6 hours fragmented refresh path	60-90 minutes controlled refresh chain
Production release traceability	Fragmented approvals across email/spreadsheet s	100% tracked through gated QA and sign-off workflow
Data definition disputes across finance and actuarial consumers	Frequent period-end measure reinterpretation	Reduced materially through centralized semantic measures and documented KPI ownership

Role and Responsibilities

- Owning the target-state architecture across ingestion, storage, semantic serving, reporting, observability, and release governance.

- Defining canonical domain boundaries, source ownership expectations, and escalation protocols for ETL defects, semantic drift, and reconciliation failures.

- Driving design choices across SSIS, SQL Server, SSAS Tabular, and Power BI based on determinism, supportability, and auditability.

- Establishing production-readiness gates, QA certification controls, and sign-off criteria for CFO, actuarial, and statutory reporting paths.

- Governing cross-functional execution across data engineering, BI engineering, QA, business stakeholders, and vendors.

- Holding budget and tooling authority for architectural choices and platform investments.

- Leading issue triage for period-close defects, late-arriving data, failed processing chains, and measure disputes.

- Translating executive objectives into practical architecture roadmaps and release outcomes.

Challenges and Resolutions

The hard parts of this program had less to do with technical novelty and more to do with financial control, semantic ambiguity, legacy integration, and release accountability.

Challenge	Why It Was Hard	Resolution
Inconsistent metric definitions across finance, actuarial, and operations	The same metric names were derived differently across teams, making reporting non-reproducible	Introduced a canonical semantic layer in SSAS with governed KPI ownership, conformance rules, and release-controlled measure definitions.

Late-arriving or partial source extracts	Reporting windows were vulnerable to upstream delays, especially around close periods.	Implemented control tables, dependency-aware scheduling, restartable ETL stages, and release holds for incomplete data conditions.
Manual reconciliation overload	Analysts spent large portions of cycle time stitching mismatches outside the platform.	Added rule-based validation, exception tables, auto-classified discrepancy buckets, and traceable remediation workflows.

Cube processing and BI refresh fragility	Downstream report consumers saw inconsistency when semantic and dataset refresh dependencies were not coordinated.	Introduced partition-aware processing order, refresh orchestration, and QA certification before report release.
Production sign-off ambiguity	Technical teams and business stakeholders lacked a formal release authority model.	Defined sign-off boundaries across QA, finance control owners, actuarial consumers, and audit reviewers before production publication.
Tooling sprawl and vendor-led fragmentation	Independent solutioning patterns increased operational entropy and support risk.	Centralized architecture authority and standardized the delivery operating model around a single governed platform pattern.

Failure Handling and Reliability Approach

The reliability model included:

- Package-level retry with bounded backoff for transient connectivity or extract issues.
- Idempotent stage publishing so reruns do not duplicate facts or corrupt snapshots.
- Reject tables and validation stores for failed conformance or key resolution cases.
- Partition-scoped semantic processing to reduce the blast radius of data issues.
- Run-level lineage capture, correlation IDs, and audit event persistence across ETL, warehouse publish, model processing, and reporting release boundaries.

- Controlled degraded mode where non-critical dashboards can be held back without releasing uncertified data.

Design Trade-offs

- Curated SQL warehouse conformance over direct source-to-report federation, because reproducibility mattered more than quick iteration.

- SSAS Tabular semantic centralization over report-local measure logic, because KPI consistency had to survive across CFO, actuarial, and audit views.

- Batch-oriented controlled refresh over pseudo-real-time complexity, because certified correctness mattered more than second-level freshness.

- Azure-native deployment alignment over cross-cloud abstraction, because the Microsoft stack already defined the most supportable baseline.

- Release gates and sign-off checkpoints over autonomous BI publication, because regulated reporting cannot rely on uncontrolled self-service semantics.

Power BI Dashboards

The platform was designed to support a clear set of operational and executive dashboards.

1. **CFO Executive Financial Intelligence Dashboard**

   Users: CFO, Finance Controller, Executive Leadership.

   Refresh Interval: Daily during close periods, otherwise business-day refresh.

   Purpose: Enterprise financial monitoring across premium, claims, loss ratio, reserve position, commissions, and legal-entity performance.

   Visuals:

   - KPI cards for Gross Written Premium, Earned Premium, Incurred Loss, Paid Loss, Combined Ratio, Expense Ratio, Commission Ratio, and Outstanding Reserve Balance.

- Variance cards for Actual vs Budget, Actual vs Prior Year, and Actual vs Forecast.

- Waterfall chart from Gross Written Premium to Underwriting Margin.

- Trend lines for premium, incurred loss, and combined ratio.

- Matrix heatmap by Legal Entity and Line of Business.

- Decomposition tree for drivers of loss ratio.

- Drillthrough page for entity-level bridge and product performance.

2. Chief Actuary Reserve And Claims Dashboard

Users: Chief Actuary, Reserving Team, Claims Analytics leads.

Refresh: Daily during reserving cycles; otherwise weekly.

Purpose: Reserve sufficiency, incurred-versus-paid development, severity/frequency trends, and reserve movement analysis.

Visuals:

- KPI cards for Case Reserve, IBNR Reserve, Total Technical Reserve, Reserve Release/Strengthening, Paid-to-Incurred Ratio, Average Claim Severity, and Claim Frequency.

- Triangle-style matrices for Accident Period vs Development Period.

- Trend lines for reserve movement, open claims, and settlement duration.

- Scatter plot for severity vs frequency.

- Small-multiple line charts for paid vs incurred development.

- Stacked bar chart for reserve composition.

- Drillthrough page for claim cohort analytics.

3. Statutory And Audit Readiness Dashboard

Users: Statutory Reporting Lead, Internal Audit, External Audit coordination teams.

Refresh: Every release cycle and during pre-filing windows.

Purpose: Measure certification, source completeness, reconciliation status, exception backlog, and evidence traceability.

Visuals:

- KPI cards for source feeds expected vs received, reconciliations passed, open high-severity exceptions, measures pending sign-off, and reports certified for release.

- Status matrix across extract completeness, conformance, reconciliation, semantic validation, business approval, and audit readiness.

- Bar chart for exception count by severity, source, and aging.

- Timeline or Gantt view for cycle milestones.

- Evidence pack table with lineage reference, run ID, model version, approver, and timestamp.

- Drillthrough page for reconciliation failures.

4. **Finance Operations And Close Control Dashboard**

Users: Finance Operations, Data Control Office, QA leads.

Refresh: Intra-day during close, daily otherwise.

Purpose: Operational control of pipelines, close dependencies, exception remediation, and report-release readiness.

Visuals:

- KPI cards for ETL jobs completed, ETL jobs failed, average load duration, warehouse publish status, SSAS processing status, and Power BI refresh success rate.

- Funnel from source extracts received to report released.

- Bar chart for load duration by source and package group.

- Queue table for open exception items.

- Trend line for failure rate and remediation time.

- Dependency map for close-critical datasets and dashboards.

5. Underwriting And Portfolio Performance Dashboard

Users: Insurance Operations Leadership, Product Heads, Finance Business Partners.

Refresh: Daily or weekly depending on source availability.

Purpose: Product-line performance, channel economics, retention, exposure insight, and profitability analysis.

Visuals:

- KPI cards for Written Premium, Earned Premium, Policy Count, Renewal Rate, Loss Ratio, Commission Ratio, Expense Ratio, and Underwriting Margin.

- Ranking chart for top and bottom product families.

- Trend lines for premium, retention, and loss ratio.

- Map visual only where geographic exposure is meaningful and source quality supports it.

- Stacked bar chart for channel mix.

- Matrix for product family, broker/channel, and profitability metrics.

6. Commission And Distribution Dashboard

Users: Distribution Finance, Sales Operations, CFO office.

Refresh: Weekly or per cycle.

Purpose: Distribution cost transparency and channel profitability analysis.

Visuals:

- KPI cards for Total Commission Paid, Commission Ratio, Agency Commission, Broker Commission, and Net Premium After Distribution Cost.

- Waterfall from written premium to net retained premium.

- Bar chart for brokers and agencies by premium versus payout.

- Scatter plot for premium volume vs commission ratio.

- Trend line for monthly commission cost.

- Tabular detail page for settlement, clawback, adjustment, and variance records.